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Learning-assisted InSAR DEM Enhancement for High-Resolution, Terrain-Aware Hydrologic Digital Twins

Muhammad Farhan Humayun[1], Getnet Demil[2], Tomi Westerlund[1], Mourad Oussalah[2], and Jukka Heikkonen[1]
[1]Department of Computing, University of Turku, Finland
[2]Center for Machine Vision and Signal Analysis, University of Oulu, Finland

Accurate Digital Elevation Models (DEMs) at high spatial resolution are a critical prerequisite for terrain-aware hydrologic Digital Twins, where topographic errors directly compromise flow routing, inundation mapping, and hazard prediction. Within the scope of hydrologic digitization, improving the reliability of DEMs derived from spaceborne Interferometric Synthetic Aperture Radar (InSAR) remains a key challenge.

InSAR is a widely used technique for DEM generation, which exploits phase differences and coherence information from two or more SAR acquisitions. While spaceborne InSAR enables large-scale and weather-independent observations, its performance is strongly constrained by sensor geometry, temporal and perpendicular baselines, and surface dynamics. In particular, coherence degradation caused by vegetation cover, soil moisture variability, atmospheric effects, and the presence of wetlands or water bodies leads to noisy interferograms and reduced DEM accuracy in hydrologically relevant environments.

This study investigates a learning-assisted InSAR framework to enhance interferometric data quality and mitigate coherence-related limitations in SAR-derived DEMs. Deep learning based generative and representation-learning models, including diffusion models, generative adversarial networks, and variational autoencoders, are evaluated to support coherence enhancement and artifact suppression in SAR image pairs. The learning components are integrated with established InSAR processing pipelines to improve interferometric phase stability and DEM quality without compromising the physical consistency of the interferometric observables.

Our methodology leverages daily repeat-track ICEYE and bistatic TerraSAR-X/TanDEM-X satellite acquisitions, with high-resolution reference DEMs from the National Land Survey of Finland enabling robust validation beyond open global DEM products. Initial experiments using Sentinel-1 image pairs show consistent improvements in interferogram quality and spatial coherence patterns relative to baseline InSAR processing, particularly in vegetated and mixed land-cover areas affected by decorrelation. Quantitative aspect of the methodology focuses on improvements in interferogram quality, elevation accuracy, and uncertainty patterns relevant for hydrologic Digital Twin applications. The workflow is designed for scalability across different SAR sensor configurations.

By addressing coherence limitations through the integration of physics-aware deep learning with InSAR pipelines, this work aims to enable more reliable, high-resolution DEM generation for terrain-sensitive hydrologic Digital Twins within the Digital Waters (DIWA) framework.